

instance, if it is a pluggable connector, check how hard it is to plug it in and plug it out, whether the locking arrangement is proper or not, and whether the polarisation is clearly mentioned or not.

When pulling out the connector, especially a PCB-to-wire connector, the physical design should also be taken into consideration. If there are any sharp edges in the connector, it can hurt fingers while pulling-out or pushing-in.

Before finalising a connector, Mahendra Karandikar prefers to get some samples and cut them open to see their internal design. Most connectors are made of brass, generally, with tinplating, chrome-plating, or zinc-plating. The other factors he checks for include surface area for contact, design of the spring action inside a connector (Is it robust enough to last long after repeated connections?), and current capacity ratings.

Consumer-domain perspective

Initially, there used to be many different types of USBs such as micro-USB Type-B-port, and Type-C port. There were separate connectors for USB 3.0 and USB 2.0. The socket for Type-A was different. It was a complicated scenario.

But now, all of that has collapsed into Type-C, which is 'idiot-proof.' It has a simple interface, which can be plugged in any direction. It is also compatible with USB 2.0 and USB 3.0, while supporting both power delivery and data connectivity.

"Type-C has completely revolutionised the way we connect." You do not need a separate power port and data port. You can have a single port connected with both of those. Soon HDMI and Ethernet cable will go away and will be replaced with Type-C," says Rajanikant Mohan, director-product management, business development and marketing, Mindtree.

Will Type-C replace HDMI and Ethernet cables?

"Absolutely," agrees Abhishek Satish, co-founder and CTO, Vicara. Many smartphones earlier came with micro-USB that made these cables popular among consumers. But what if today your microcontroller did not support micro-USB? Then you will need to



manage your firmware updates and wired-in applications in a completely different way, making things really complicated.

However, Type-C has changed everything. "Nowadays, everything is supported by Type-C. This was difficult to achieve back then when all electronic devices such as laptops used to come with their own separate adapters. Now it is a lot easier for the consumer to have a single device charger/cable that connects with all. So, I definitely see Type-C being the future, especially since it supports tremendous amounts of bandwidths," he says.

Stating his experience as a MacBook user that has a Type-C port, he finds it extremely convenient to charge his laptop with a power bank (which also has a Type-C port) while on the go. Consumers find it much easier to have a single Type-C cable that connects with everything.

This is an interesting point, given that the traditional USB 3.0 port or USB 2.0 port earlier provided 4.5W of power (9mA of current at 5V), which was not enough. With Type-C, one can go up to 5A over 20V, which allows you to charge your laptop, run a monitor, and do much more - using just a single port.

Designing a PCB that can support power coming in from Type-C is not a trivial task. A controller tells the power circuitry about the power rating of the device that the cable is connected to regarding how much current is going to get driven out of the power lines (connected to the USB

port). From that perspective, several reference designs are being created by semicon companies, which need to be studied if you are trying to get your device charged and powered from a USB Type-C port.

Non-consumer domain

In the consumer domain, the way power is distributed is getting standardised to USB Type-C. However, the same is not happening from an industrial point of view as there is no specific standard for connectivity. "In the case of consumer electronics, you have to collect power and data together through USB connection; the same cable is used for charging a phone and data extraction. As far as industrial communication is concerned, there are a variety of communication standards," says Mahendra Karandikar.

PLC manufacturers have their own communication standards. Siemens is coming up with PROFIBUS over RS 485, PROFIBUS over Ethernet, etc. CAN bus protocol, MOD bus protocol, and RS 485 protocol are also used in industry. And every standard has its own connector design. For Ethernet, there is RJ45 while for PROFIBUS, generally, it is a 9-pin D-Sub connector. It is very rare in industry to see power and signal being delivered through the same connector cable.

"Standardisation in the consumer and IT sector is a different ball game altogether. **Industrial designs are slightly flexible and there is a lot of freedom to choose the connectors.** An organisation standardises as per their variety of